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Optimizing Unconventional Reservoir Development: Efficient Multi-Well Performance Modeling Using the Boundary Element Method

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Abstract

In certain scenarios of unconventional resource development, reservoir geology enforces an irregular placement of wells and hydraulic fractures. The resulting reservoir-well architecture and the attendant subsurface flow pattern may thus pose formidable reservoir/well performance modeling challenges. Under this circumstance, analytical solutions compromise on the representation of actual reservoir- well geometries while numerical simulation suffers extreme gridding requirements and incurs extended runtimes. This study presents a practical semi-analytical solution for modeling multi-well performance in tight reservoirs, with rigorous representation of complex well geometries and exact consideration for well interference.

The proposed solution utilizes the boundary-element method with an appropriate Green's function to model fully and partially penetrating hydraulic fractures of arbitrary azimuth. Well laterals are unrestricted in both orientation and deviation; the reservoir lateral boundaries can be arbitrary-shaped or of infinite extent. Multi-well production and interference are handled using the superposition principle. We benchmarked our solution against analytical and numerical models for variable well production (driven by either rate and pressure control), as well as for regular and irregular hydraulic fractures. The results are in excellent agreement.

Verification cases highlight our solution's fit for modeling the flow behavior of fully and partially penetrating hydraulic fractures, notwithstanding the irregularity in fracture geometry. Three application examples are presented: the first evaluates completion effects on well performance and determines the optimum fracture spacing by evaluating incremental production; the second example examines interference from an infill well and its impact on the parent well's performance; the third example demonstrates application in history matching and long-term production forecast. The accuracy of this semi-analytical solution makes it useful for transient analysis and for initial calibration and verification of the more complex numerical simulation. Because of its rigorous multi-well modeling capability, it can also be deployed in synchronization with and as a reflection of actual field development for reservoirs producing single-phase oil or gas.

Introduction

The gains of unconventional resource exploitation have been further enhanced by multi-well pad drilling technology through enabling reduced surface footprints, increased well density, and, ultimately, access to a larger reservoir flow area. This has made the utility of multi-fractured wells in tight reservoirs the more appealing because of the economies of scale. Although a development plan based on a regular well pattern is desirable, subsurface geology, however, occasions actual well placement (laterally and vertically) in the field. The distribution of reservoir heterogeneities may be such that irregular and/or asymmetrical well patterns best penetrate the sweet spots. For reservoir management purposes, it is imperative to understand the flow performance of multi-well production set-up, with rigorous consideration of the impact of interference. Analytical models may be used in this regard; however, they are limited to regular reservoir-well architectures. Advanced numerical models, on the other hand, are capable of representing geometrically complex well laterals and reservoir boundaries, however, they do so at the cost of gridding and runtime challenges.

In their work, [Ajani and Kelkar \(2012\)](#) provided recommendations for optimal well spacing by quantifying the effect of well interference on production performance. Examining fracture data vis-a-vis daily gas and water production data of horizontal wells producing for over five years in the Arkoma basin, they developed a method to calculate the probability of interference as a function of distance and age of neighboring wells and determine the preferential direction of interference. [Buchwalter et al. \(2012\)](#) presented a method for estimating optimal infill well location by use of production and pressures of vertical discovery wells. Their workflow involves: developing pattern models for the vertical wells, improving the reservoir description of the models through computer-assisted history matching, and using a geostatistical technique to propagate the reservoir property improvement over inter-well regions. The resulting model is then used to investigate the impact of multi-fractured horizontal well location and completion strategies.

Analyzing the flow behavior of wells participating in a multi-well-pad production, [Awada et al. \(2016\)](#) provided a diagnostic procedure for identifying interference by observing changes in pressure build-up trend as the wells are staggered on/off production.

Various mechanisms of interference are defined in their study - caused by fracture hits, adjoining hydraulic fractures, natural fractures, reservoir matrix, and gathering systems. Using numerical simulation, they history-matched the performance of an eight-well in the Horn River basin and confirmed the interference mechanism between communicating wells. [Hamdi et al. \(2021\)](#) developed a material-balance-driven semi-analytical method to analyze production data from interfering multi-fractured horizontal wells on the same pad, producing a from shale gas reservoir. They represented the reservoir matrix and the hydraulic fractures adjacent to communicating wells as individual compartments which interact via buffer zones. Solving the resulting system of ordinary differential equations using a numerical integration scheme, they obtained results which closely approximate well performance. The degree of communication is deduced from the properties of the intervening buffer zones.

In this study, we develop a rigorous semi-analytical solution for modeling the performance of and the interference between multifractured wells producing from a single-phase oil or gas reservoir. The solution is based on liquid flow; however, gas flow is handled via pseudo-variables ([Al-Hussainy et al. 1965](#), [Agarwal 1979](#)). We use the source function approach ([Gringarten and Ramey 1973](#), [Falade 1979](#), [Ozkan and Raghavan 1991](#)) to model hydraulic fractures with arbitrary azimuth and arbitrary vertical penetration in a uniform pay reservoir. Hence, complex well laterals are adequately represented. Both finite and infinite reservoirs can be simulated; boundary influence is incorporated using a boundary-element-method (BEM) technique, thus enabling our solution to accommodate arbitrary-shaped reservoirs. In terms of well control, variable rate and variable pressure flow strategies are possible. Note that the solution is in Laplace space ([Everdingen and Hurst 1949](#)); real-time flow results are obtained using the Stehfest inversion algorithm ([Stehfest 1970a](#) and [Stehfest 1970b](#)). Our solution can be put to a variety of performance modeling

applications e.g. fracture/cluster spacing studies, well spacing evaluation, calibration and verification of numerical simulation, history matching and forecasting, to mention a few. We present a number verification and application examples to demonstrate its versatility.

Mathematical Formulation

The solution put forward in this study applies to single-phase gas and single-phase liquid flow. Whereas the following mathematical development is based on the flow of liquids (slightly compressible fluids), adaption to the flow of gases (highly compressible fluids) is handled by use of pseudo-variables, including pseudo-pressure (Al-Hussainy et al. 1965) and pseudo-time (Agarwal 1979). As stated by Idorenyin and Shirif (2015, 2016) and Idorenyin (2016), the flow dynamics of a slightly compressible fluid in a porous media is governed by the following boundary-integral equation:

$$p_D(\mathbf{x}_D, s) = 2\pi\eta_D \sum_{\ell=1}^M \int_0^{t_D} \tilde{q}_{wD\ell}(\tau_D) S(\mathbf{x}_D; t_D - \tau_D) d\tau_D + \eta_D \int_0^{t_D} \int_{\Gamma} \left[G(\mathbf{x}_D, \mathbf{x}'_D; t_D - \tau_D) \nabla p_D(\mathbf{x}'_D, \tau_D) - p_D(\mathbf{x}'_D, \tau_D) \nabla G(\mathbf{x}_D, \mathbf{x}'_D; t_D - \tau_D) \right] \cdot \hat{\mathbf{n}} d\Gamma' d\tau_D \cdot \hat{\mathbf{n}} d\Gamma' \quad (1)$$

which, in Laplace space, becomes

$$\bar{p}_D(\mathbf{x}_D, s) = 2\pi\eta_D \sum_{\ell=1}^M \tilde{q}_{wD\ell}(s) \bar{S}(\mathbf{x}_D, s) + \eta_D \int_{\Gamma} \left[\bar{G}(\mathbf{x}_D, \mathbf{x}'_D; s) \nabla \bar{p}_D(\mathbf{x}'_D, s) - \bar{p}_D(\mathbf{x}'_D, s) \nabla \bar{G}(\mathbf{x}_D, \mathbf{x}'_D; s) \right] \cdot \hat{\mathbf{n}} d\Gamma' \quad (2)$$

where

$$\bar{S}(\mathbf{x}_D, s) = \int_{\Omega} \bar{G}(\mathbf{x}_D, \mathbf{x}'_D; s) d\Omega \quad (3)$$

The subscript marker "D" in the above denotes a dimensionless quantity. $p_D(x_D, s)$ is the pressure observed at point x_D in the reservoir, $S(x_D; t_D - t_d)$ is the source function with strength $q_{wD}(t_D)$ and active for the elapsed time $t_D - t_d$, $G(x_D, x'_D; t_D - t_d)$ is the corresponding Green's function, p_D is the hydraulic diffusivity of the flow system, $\mathbf{n}$ is the outward unit normal to the boundary Γ , and s is the Laplace space parameter with respect to the time variable. M is the number of sources in the flow domain Ω bounded by the surface Γ . Eq. 2 assumes that the initial reservoir pressure is uniform across the flow domain, so that $p(x, 0) = p$, implies $p_D(x_D, 0) = 0$. The dimensionless variables in Eq. 2 are defined in Appendix A.

This study focuses on fluid flow in a slab reservoir, i.e. a constant-pay flow domain with flat top and bottom boundaries. We restrict the conditions of the top and bottom boundaries to no-flow. The reservoir's lateral extent may be finite or infinite. Fig. 1 is an illustration of an arbitrary-shaped, finite, slab reservoir with three multi-fractured horizontal wells. The hydraulic fractures are vertical (some of which penetrate the pay fully and the rest partially) and are of arbitrary azimuth. The well laterals are unrestricted in orientation; they may be deviated and/or slanted. Our implementation of Eq. 2 adopts the boundary-element-method (BEM) technique. This requires discretization of the reservoir boundaries and the hydraulic fractures, as shown in Fig. 1. Any of a variety boundary element geometry may be used (including triangle, quadrilateral, hexagon, etc.); we settle for the quadrilateral element in this study because of the simplicity it affords in computation.

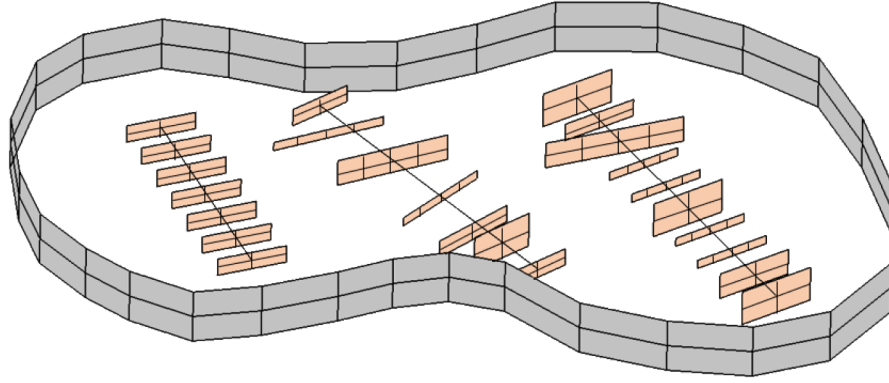


Figure 1—An arbitrary-shaped, constant-pay reservoir with discretized boundaries. For each of the wells, its hydraulic fractures are discretized into segments; a non-discretized hydraulic fracture is essentially a one-segment source

The source function we employ in our solution encapsulates no-flow top and bottom boundaries within its formulation, and the quadrilateral boundary element integrates easily with it (see Appendix B, Fig B-1). This approach also helps limit the number of elements required to represent the reservoir boundaries and, by so doing, reduces the computational overhead. We progress the solution development for finite, arbitrary-shaped reservoirs and, subsequently, modify the resulting solution for infinite reservoirs.

Finite Reservoir

We adopt the isoparametric BEM formulation (Kikani and Horne 1989, 1993, Idorenyin and Shirif 2018). The boundary quantities (pressure and flux) as well as the boundary elements are approximated with a bilinear shape function. The nodes of the shape function are described on a 2D coordinate system intrinsic to each boundary element (see Appendix C, Table C-1). By so doing, the dimensionality of the problem reduces from 3D to 2D. This is one of the notable gains of the boundary element method. To determine the values of the boundary quantities, we apply a collocation procedure over the boundary nodes, as depicted in Appendix C, Fig C-1. This generates a system of equations which is solved using a standard linear solver. The mathematical details are deferred to Appendix C. As given by Eq. C-18 in Appendix C, the Laplace-space transient response in a reservoir with M number of sources and N boundary nodes is

$$A\bar{p} + B\bar{q} + C\bar{q}_{wf} = 0 \quad (4)$$

where A and B are $N \times N$ matrices, C is an $N \times (M \times N)$ matrix, p and q are N -element vectors denoting boundary quantities, q_w is an N -element vector representing source strength, and 0 is an N -element zero vector. The equation shows, in a concise form, that the transient response in the reservoir is due to the effects of boundary pressure (p), boundary flux (q) and well flow rate (q_w). A and B are coefficients which incorporate the influence of the reservoir boundary geometry and C the source geometry.

We assume that the lateral boundary of the reservoir is a no-flow boundary. As a result, the boundary flux is

$$q = 0 \quad (5)$$

Based on this assumption, Eq. 4 therefore reduces to

$$A\bar{p} + C\bar{q}_{wf} = 0 \quad (6)$$

Note that the well flow-rate vector (q_{wf}) denotes the flow intensity of the hydraulic fracture segments of the M wells. That is,

$$\bar{q}_{wf} = [\bar{q}_{wf1}, \bar{q}_{wf2}, \dots, \bar{q}_{wfM}]^T \quad (7)$$

Given that each well I has N_{fI} hydraulic fracture segments, then there are $N + Z_I N_{fI}$ unknowns (p and q_{wf}) but only N equations exist. The remaining N_{fI} equations required to make the system well-defined are obtained by collocating at the N_{fI} hydraulic fracture segments. This yields the following additional set of equations:

$$\bar{p}_{wf} + A^w \bar{p} + C^w \bar{q}_{wf} = 0 \quad (8)$$

The pressure vector ($\bar{p}_{wf}$) denotes the pressures hydraulic fractures. That is,

$$\bar{p}_{wf} = [p_{wf1} \ p_{wf2} \ \dots \ p_{wfM}]^T \quad (9)$$

Eq. 8 introduces $Z_I N_{fI}$ additional unknowns ($\bar{p}_{wf}$) making a total of $N + 2 Z_I N_{fI}$ unknowns for the system, but there are only $N + Z_I N_{fI}$ equations. So, the system of equations is still ill-defined. We assume that, for each of the M wells, its hydraulic fractures flow at the same pressure, p_w , as its infinitely conductive lateral. Based on this assumption, the number of unknowns reduces to $N + M + Z_I N_{fI}$. This means we now require M more equations to make the system well-defined. These new equations are obtained from each well's production rate. We require that the summation of the fracture-segment flow rates for each well be equal to the well's production rate. That is,

$$q_{w\ell}(t) = \sum_{i=1}^{N_{f\ell}} q_{wf,i}(t) \quad (\ell = 1, 2, \dots, M) \quad (10)$$

Combining Eqs. 6 through 10, the system of equations to solve becomes

$$\begin{bmatrix} A & C & 0 \\ A^w & C^w & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \bar{p} \\ \bar{q}_{wf} \\ \bar{p}_w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \bar{q}_w \end{bmatrix} \quad (11)$$

By solving Eq. 11, the wellbore pressure and fracture-segment flow rates for each well are obtained as part of the solution, albeit in Laplace space. The real-time pressures and rates are obtained by inverting the computed response using a numerical Laplace inversion procedure. In this paper, we use Stehfest's inversion algorithm (Stehfest 1970a and Stehfest 1970b).

Infinite Reservoir

Eq. 11 applies to a finite reservoir. If, however, the reservoir is unbounded laterally, the transient response in the flow domain still respects Eq. 11 but without the boundary terms. So, for an infinite reservoir with M multi-fractured horizontal wells, the system of equations simplifies as

$$\begin{bmatrix} C^w & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \bar{q}_{wf} \\ \bar{p}_w \end{bmatrix} = \begin{bmatrix} 0 \\ \bar{q}_w \end{bmatrix} \quad (12)$$

Well Controls

Wells are operated either based on flow rate control or pressure control. Eq. 11 and Eq. 12 developed in the previous section only apply to rate-controlled wells, for finite and infinite reservoirs respectively. So, we will not re-state the solution for rate-controlled wells here. However, observe that the well flow-rate vector (q_w) appears on the right-hand side of each equation because it is specified for this type of problem.

In the rest of this section, we present the development of the pressure-control solution for finite reservoirs first, and then modify the resulting solution for infinite reservoirs.

Finite Reservoir

For wells operated at specified bottom-hole pressure (p_w), the solution follows the same line of development as in the previous section. The boundary-collocated equation given by Eq. 6 also holds for this reservoir-well setup. However, since well pressures are known, the well-collocated equation given by Eq. 15 simplifies as

$$A^w \bar{p} + C^w \bar{q}_{wf} = -\bar{p}_w \quad (13)$$

Note that Eq. 13 assumes the hydraulic fractures are at the same pressure, p_w , as their infinitely conductive host well. Combining Eqs. 6 and 13, the system of equations for pressure-controlled wells in a finite reservoir is

$$\begin{bmatrix} A & C \\ A^w & C^w \end{bmatrix} \begin{bmatrix} \bar{p} \\ \bar{q}_{wf} \end{bmatrix} = \begin{bmatrix} 0 \\ -\bar{p}_w \end{bmatrix} \quad (14)$$

Infinite Reservoir

If the wells are in a laterally unbounded reservoir, the solution is the same as in Eq. 14 but with the boundary terms removed. So, the pressure-control solution for infinite reservoirs is

$$[C^w] [\bar{q}_{wf}] = [-\bar{p}_w] \quad (15)$$

Source Geometry

The elements of the matrices A , C , A^w , and C^w appearing in the solutions above involve integration of the point source function or its derivative given Eq. B-8 and Eq. B-9 in Appendix B. We integrate these equations over the source geometry using Gauss-Legendre quadrature (Abramowitz and Stegun 1965), a numerical integration technique. Integral of the first term is easily achieved, but that of the second term is tedious because of the infinite series. To mitigate this computational expense, we restrict the sources (hydraulic fractures and boundary elements) to a strictly vertical geometry. This reduces the tedium of integration of the second term, as the integral is decomposed into a "z" component and an "x-y" component, where each is handled independently. Note that for a fully penetrating source, the integral of the second term in both Eq. B-8 and Eq. B-9 vanishes - which physically interpretes as zero geometric skin. Thus, the source integral is further simplified, hence saving additional computational overhead.

Verification of Solution

Along the lines of verification, we set up various flow scenarios to capture different aspects of our solution. The result obtained in each scenario is compared with that from an equivalent analytical and/or numerical model constructed in KAPPA-Workstation (KAPPA Engineering 2025), an industry-popular dynamic data analysis software package.

Fully vs. Partially Penetrating Hydraulic Fractures in an Infinite Reservoir

This verification example sets up a multi-fractured horizontal well (MFHW) in a constant-pay, laterally infinite, gas reservoir with no-flow top and bottom boundaries. Two scenarios are considered: the first is driven by a fully-penetrating MFHW well and the second by a partially-penetrating MFHW well. Hydraulic fractures in each scenario are isometric. The conceptual representation of this reservoir-well architecture is depicted diagrammatically in Fig. 2. Table 1 outlines the reservoir properties, gas properties, and well specification. Gas critical properties are computed using Sutton (1985) correlation, gas deviation factor using Dranchuk & Aboukassam (1975) correlation, and gas viscosity using Lee et al (1966) correlation.

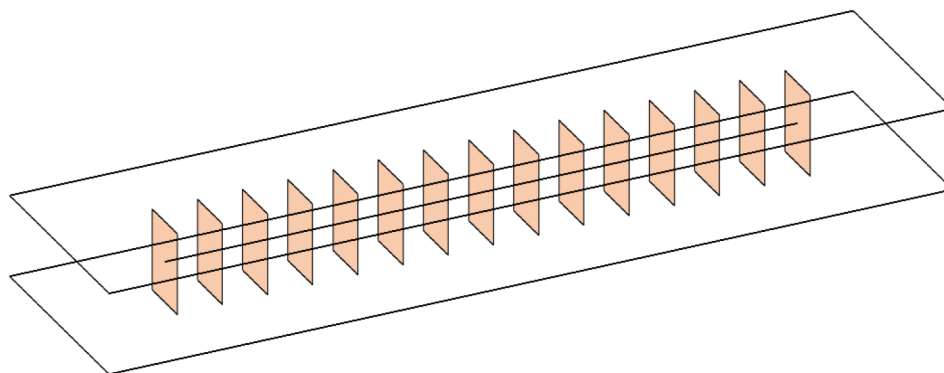


Figure 2—Perspective view of MFHW in constant-pay, laterally infinite reservoir with no-flow top and bottom boundaries

Table 1—Reservoir properties, gas properties, and well specification for fully- and partially-penetrating MFHW wells

<i>Reservoir Properties</i>	k (md)	0.001	
	(%)	12	
	c_f (psi ⁻¹)	3.0E-6	
	h (ft)	200	
<i>Gas Properties</i>	p_i (psi)	5000	
	T (°F)	120	
	γ_g	0.75	
<i>Well Specification</i>		Fully-Penetrating	Partially-Penetrating
	r_w (ft)	0.25	0.25
	L_w (ft)	3000	3000
	z_w (ft)	100	100
	N_f	15	15
	x_f (ft)	250	250
	h_f (ft)	200	100

In each scenario, production is controlled by variable bottomhole flowing pressure over a period of one year as depicted in Fig. 3; the flow rate response is as displayed in the figure. Dependence of gas properties on pressure is handled through use of pseudo-pressure and pseudo-time while the hydraulic fractures are treated as uniform-flux sources. Notice the excellent agreement between our solution and KAPPA's analytical model. The figure also shows that the fully-penetrating MFHW well outperforms the partially-penetrating MFHW well over each pressure drop because the former has a larger contacted reservoir area. However, the production rate of the fully- penetrating MFHW declines faster than and approaches that of the partially-penetrating MFHW well during each pressure step.

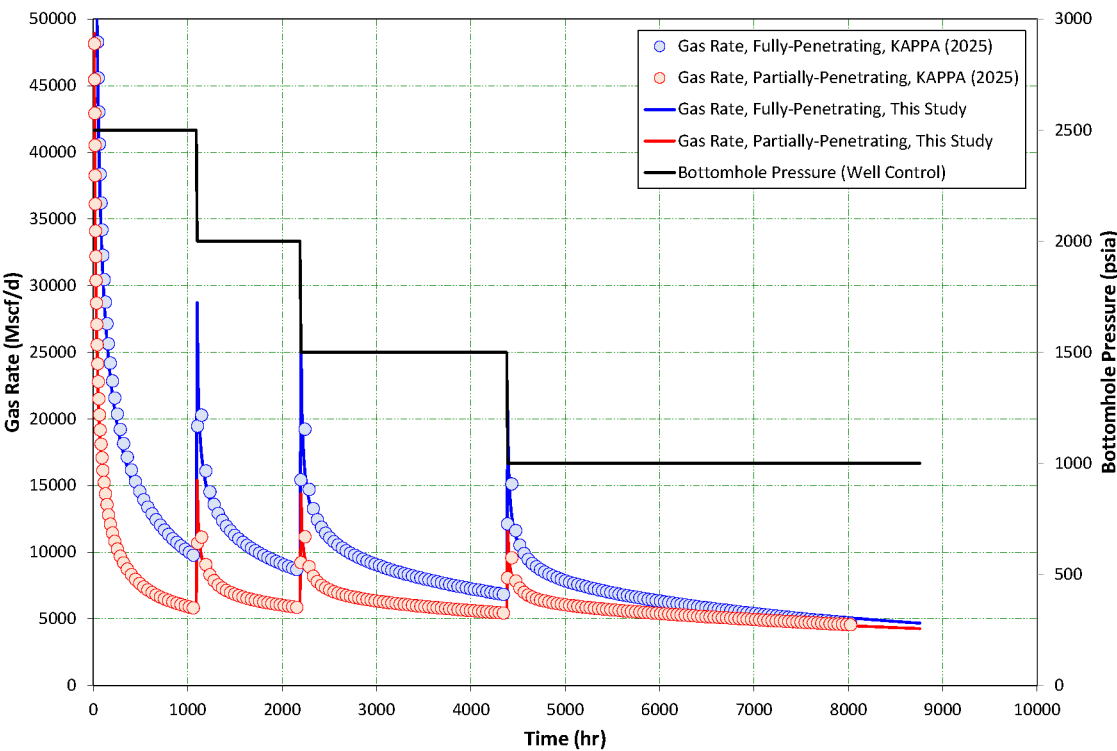


Figure 3—Gas production rate at the well due to variable bottomhole flowing pressure control

Uniform-Flux vs. Infinite-Conductivity Hydraulic Fractures in a Rectangular-Shaped Reservoir

A further verification is to assess how our solution handles uniform-flux and infinite-conductivity fractures. For the purpose of close examination, this example focuses on one hydraulic fracture set up as a fully-penetrating source in a rectangular-shaped oil reservoir, as shown in Fig. 4. An infinite- conductivity fracture is handled as a discretized source while a uniform-flux fracture is not discretized. Discretization is permitted along the length and width of the source. In addition, the reservoir boundaries are discretized. The reservoir properties, oil properties, and well specification are provided in Table 2.

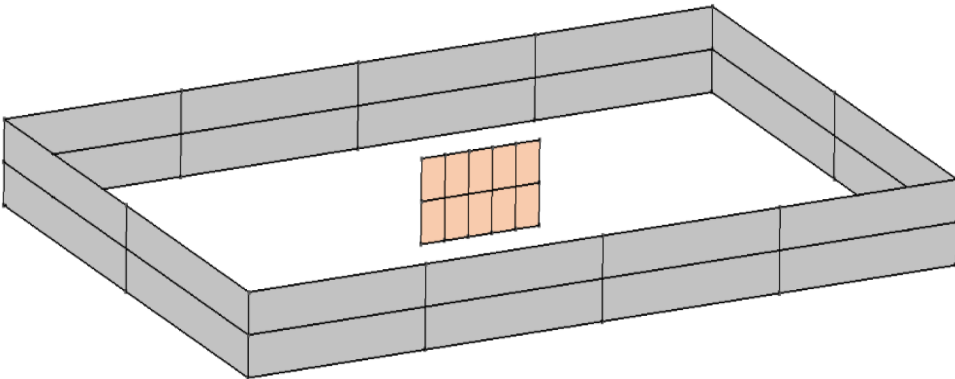


Figure 4—A Discretized hydraulic fracture representing an infinite-conductivity source in a rectangular-shaped reservoir with discretized boundaries; a uniform-flux hydraulic fracture does not require discretization

Table 2—Reservoir properties, oil properties, and well specification for uniform-flux and infinite-conductivity fractures

Reservoir Properties	k (md)	0.45
	(%)	10
	c_f (psi ⁻¹)	3.0E-6
	x_e (ft)	3000

	y_e (ft)	2000	
	h (ft)	100	
	Discretization ($N_x \times N_y \times N_z$)	$2 \times 4 \times 1$	
Oil Properties	p_i (psi)	5000	
	B (bbl/stb)	1.2	
	μ_o (cp)	0.25	
Well Specification		Uniform-Flux	Infinite-Conductivity
	r_w (ft)	0.25	0.25
	z_w (ft)	50	50
	N_f	1	1
	x_f (ft)	250	250
	h_f (ft)	100	100
	Discretization ($N_l \times N_w$)	1×1	10×1

The well is produced at a rate of 1,000 stb/d (see Fig. 5) for both the uniform-flux and infinite-conductivity fracture cases. Fig. 5 shows an excellent agreement in bottomhole pressure response between our solution and KAPPA's analytical model. The uniform-flux fracture case assumes a uniform production across the fracture area while the infinite-conductivity fracture case enforces zero pressure drop across the fracture segments. This translates to enhanced productivity for the latter case - which, given the same flow rate, manifests as a lower pressure decline than for the former.

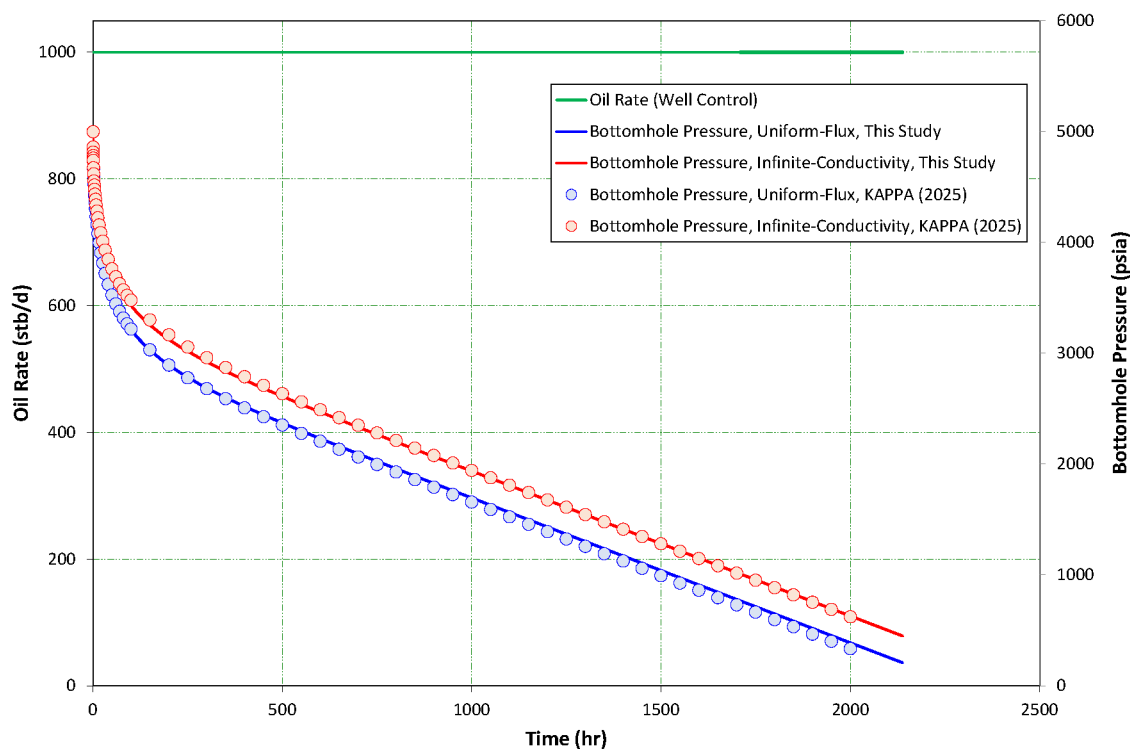


Figure 5—Bottomhole pressure response at the well due to constant flow-rate control

The rest of this section examines the impact of source discretization on our solution's accuracy for infinite-conductivity fractures.

Three types of discretization schemes are pursued based on fracture dimensions - including lengthwise discretization, widthwise discretization, and bidirectional discretization. Using the exact same set-up above, we compute the pressure response for six levels of discretization, as will be described in the results below. The results for a fully-penetrating vertical fracture are presented in Fig. 6 in terms of mean absolute error (MAE). It is seen that the lengthwise discretization and the bidirectional discretization improve in accuracy exponentially as the number of segments increases. This is because these discretization types capture realistic flux distribution in the fracture: with extended flow time, flux increases from the central fracture segment to the outermost segments at the fracture tips. The widthwise discretization lacks this form of flow development. It instead maintains equal flux contribution from the fracture centre to the fracture tips. As a result, the overall response is similar to that of a uniform-flux fracture model (represented by the 1×1 discretization) irrespective of the number of segments. In light of the above, the lengthwise discretization suffices for modeling fully-penetrating, infinite-conductivity fractures.

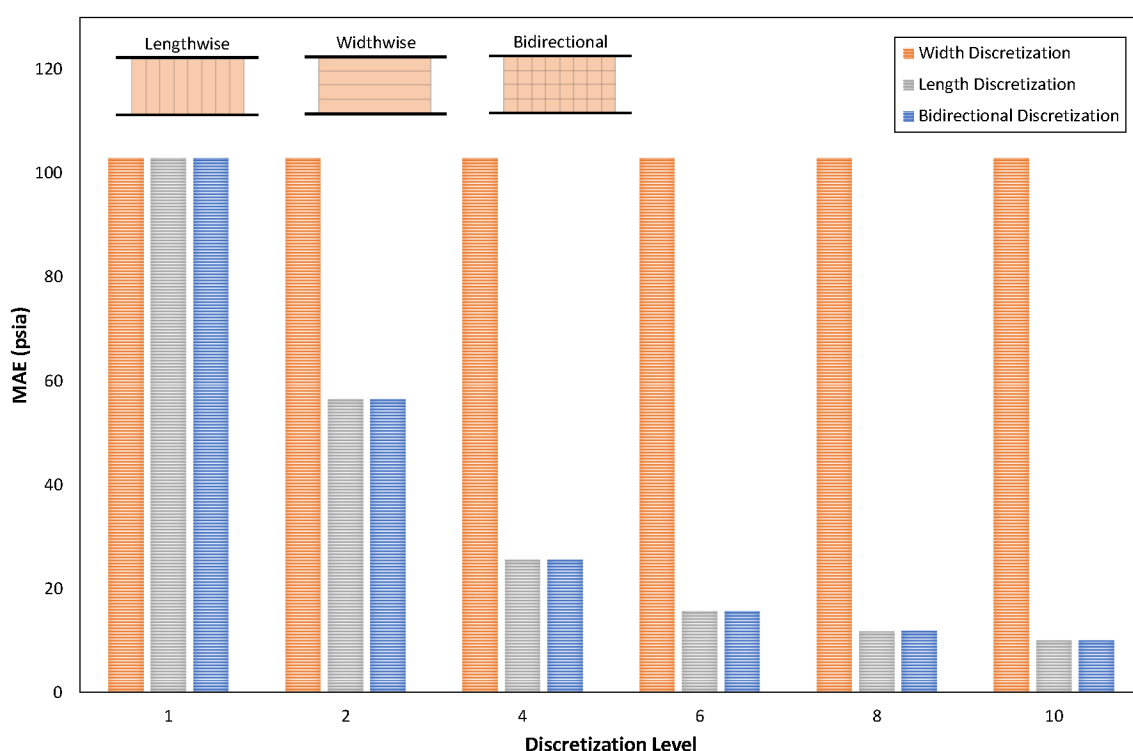


Figure 6—Results for fully-penetrating vertical hydraulic fracture model comparing lengthwise, widthwise, and bidirectional discretization in terms of MAE error analysis

We repeat the above test for a partially-penetrating vertical fracture. The reservoir properties, oil properties, and well specification are the same as in Table 2. The difference here is that the fracture height (h_f) is 50 ft. The results are presented in Fig. 7 in terms of mean absolute error (MAE). First, it is worth noting that, since the fracture is partially penetrating, the flow is three-dimensional - both horizontal and vertical flow geometries will develop between the reservoir and fracture. The relative contribution of each flow geometry is dictated by the magnitude of clearance of the fracture from the top and bottom boundaries. The hydraulic fracture considered in this example has 25 ft clearance from both the top and bottom boundaries. As such, vertical flow is significant as does horizontal flow in the reservoir-well dynamics. Fig 7 reveals that both the lengthwise and the widthwise discretization schemes are not sufficient to accurately describe the well's transient response. The bidirectional discretization shows superior performance, as demonstrated by its high rate of error reduction. This informs the choice of bidirectional discretization for modeling partially-penetrating, infinite-conductivity fractures.

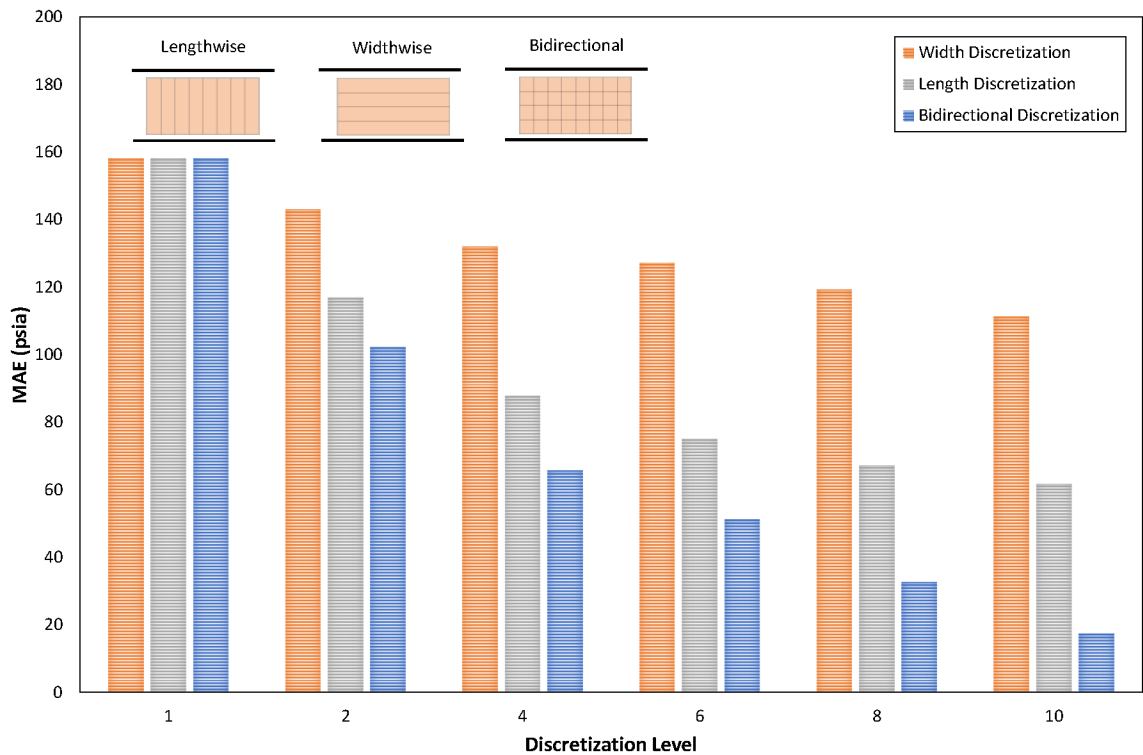


Figure 7—Results for partially-penetrating vertical hydraulic fracture model comparing lengthwise, widthwise, and bidirectional discretization in terms of MAE error analysis

Irregular Hydraulic Fractures in an Arbitrary-Shaped Reservoir

In this last verification case, we set up a multi-fractured horizontal well in an arbitrary-shaped, constant-pay, gas reservoir with no-flow top and bottom boundaries. The hydraulic fractures are irregular: they have different half-lengths, arbitrary azimuths, and fully penetrating. The reservoir-well architecture is as shown in Fig. 8. Reservoir properties, gas properties, and well specification are detailed in Table 3. Gas critical properties are computed using Sutton (1985) correlation, gas deviation factor using Dranchuk & Aboukassem (1975) correlation, and gas viscosity using Lee et al (1966) correlation.

Table 3—Reservoir properties, oil properties, and well specification for irregular-fracture MFWH well in an arbitrary-shaped reservoir

Reservoir Properties	k (md)	0.001
	(%)	12
	c_f (psi ⁻¹)	3.0E-6
	h (ft)	100
	Discretization ($N_{xy} \times N_z$)	24×1
Gas Properties	Same as Table 1	
Well Specification	r_w (ft)	0.25
	z_w (ft)	50
	L_w (ft)	3000
	N_f	15
	x_f (ft)	Varying
	h_f (ft)	100

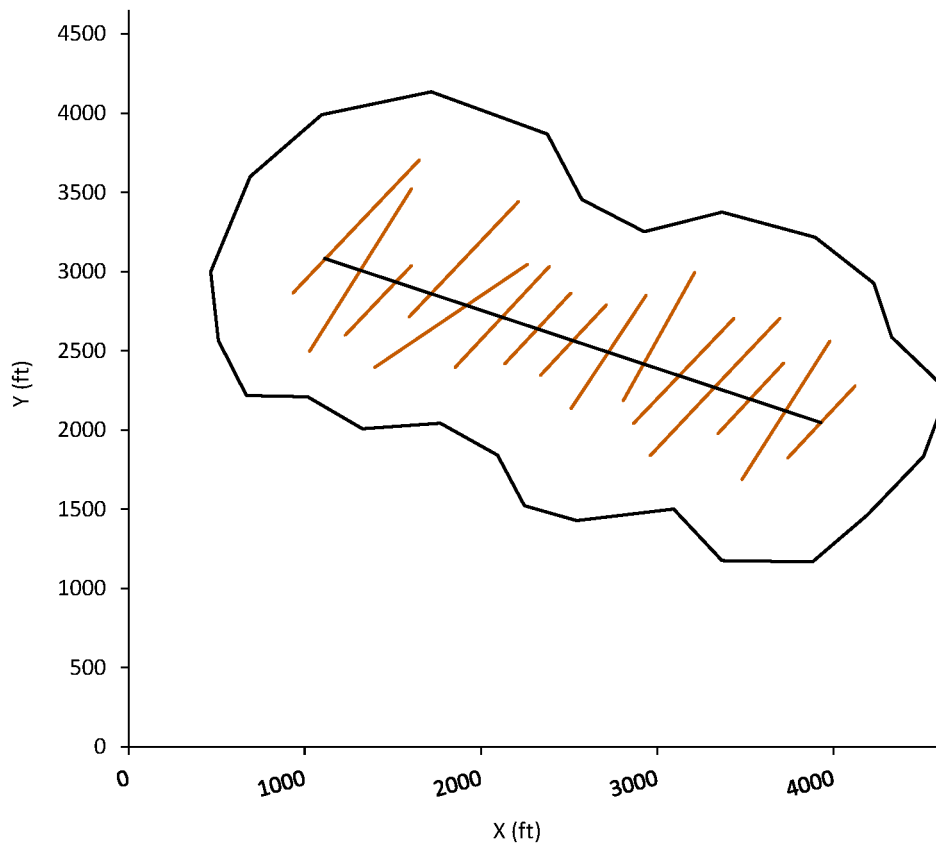


Figure 8—Plan view of an irregular-fracture MFHW well in an arbitrary-shaped, constant-pay reservoir with no-flow top and bottom boundaries

Fig. 9 shows the gas rate control imposed on the well and its bottomhole pressure response plotted for both our solution and an equivalent numerical model set up in KAPPA-Workstation. Dependence of gas properties on pressure is handled through use of pseudo-pressure and pseudo-time while the hydraulic fractures are treated as infinite-conductivity sources. The agreement in results is excellent, confirming the robustness of our semi-analytical solution.

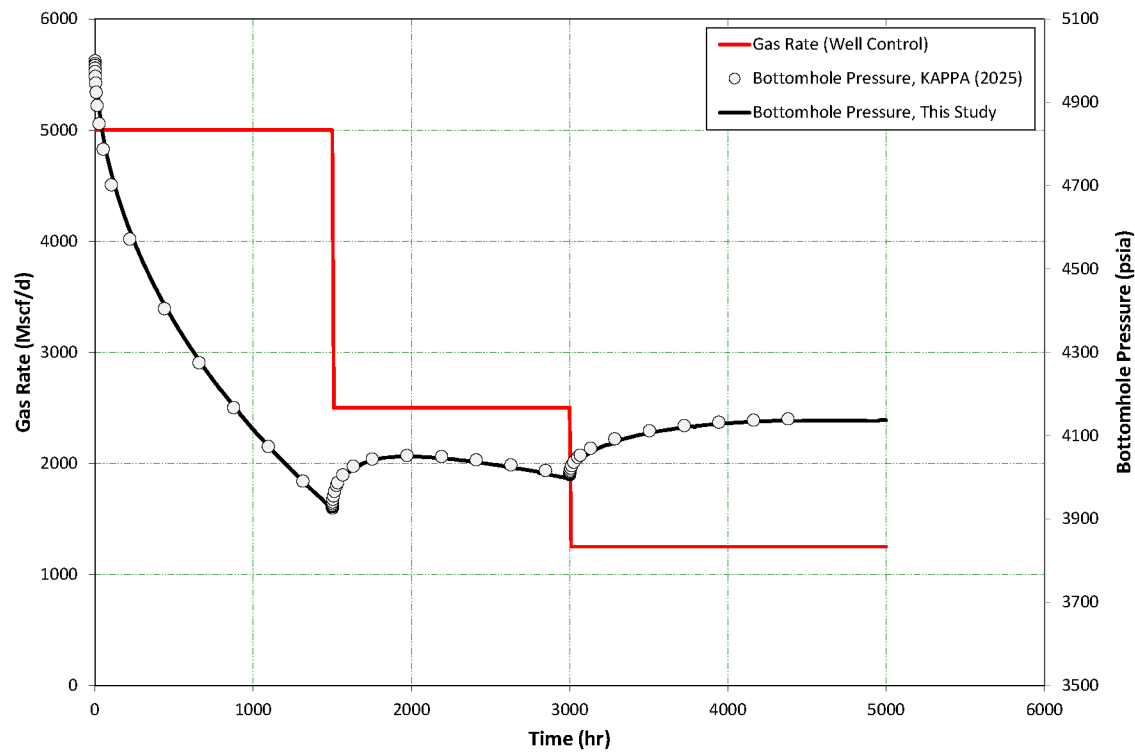


Figure 9—Bottomhole pressure at the irregular-fracture MFHW well due to variable flow-rate control

Application Examples

In this section, we showcase three applications examples, amongst other possibilities, of our semi-analytical solution. The first example evaluates completion effects on well performance and determines the optimum fracture spacing through evaluation of incremental production; the second example examines interference from an infill well and its impact on the parent well's performance; the third example demonstrates application in history matching and long-term production forecast.

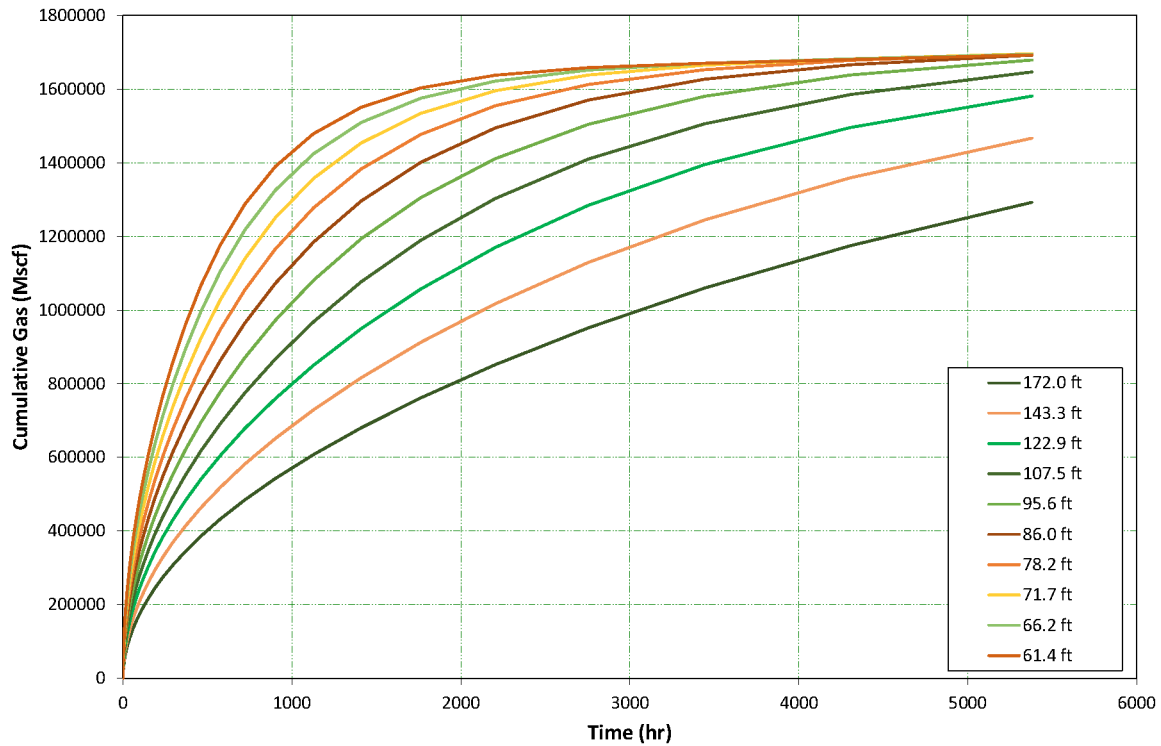
Example 1: Fracture Spacing

This example demonstrates the sensitivity of multi-fractured horizontal well performance to fracture spacing. The well in question is operated at a constant bottomhole pressure of 1,000 psia in a dry gas reservoir for 7.5 months such that reservoir boundary effects are yet to set in. Table 4 details the reservoir and gas properties as well as the well specification. Fracture spacing is varied and the cumulative gas production is monitored over the flow duration.

Table 4—Reservoir properties, oil properties, and well specification for Example 1

Reservoir Properties	Same as Table 2	
Gas Properties	Same as Table 1	
Well Specification	r_w (ft)	0.25
	L_w (ft)	2580
	z_w (ft)	50
	N_f	Varying
	x_f (ft)	250
	h_f (ft)	100

Fig. 10(a) plots the cumulative gas production for ten different fracture spacings (from 61.4 ft to 172.0 ft). It is immediately obvious that, although tighter fracture spacings accelerate resource recovery, they, however, peak early, at which point incremental production is marginal. In Fig. 10(b), we plot terminal cumulative production versus fracture spacing for all the sensitivity cases. Also displayed on the plot is the incremental gas production of each fracture spacing. Incremental production is computed as the difference in production between two proximal fracture-spacing specifications. Observe the trend of incremental production in Fig. 10(b) shows that production gains are insignificant beyond a fracture spacing of 71.7 ft. This spacing value depicts an operational optimum; however, final decisions must factor in economics to arrive at a robust completion program.



(a)

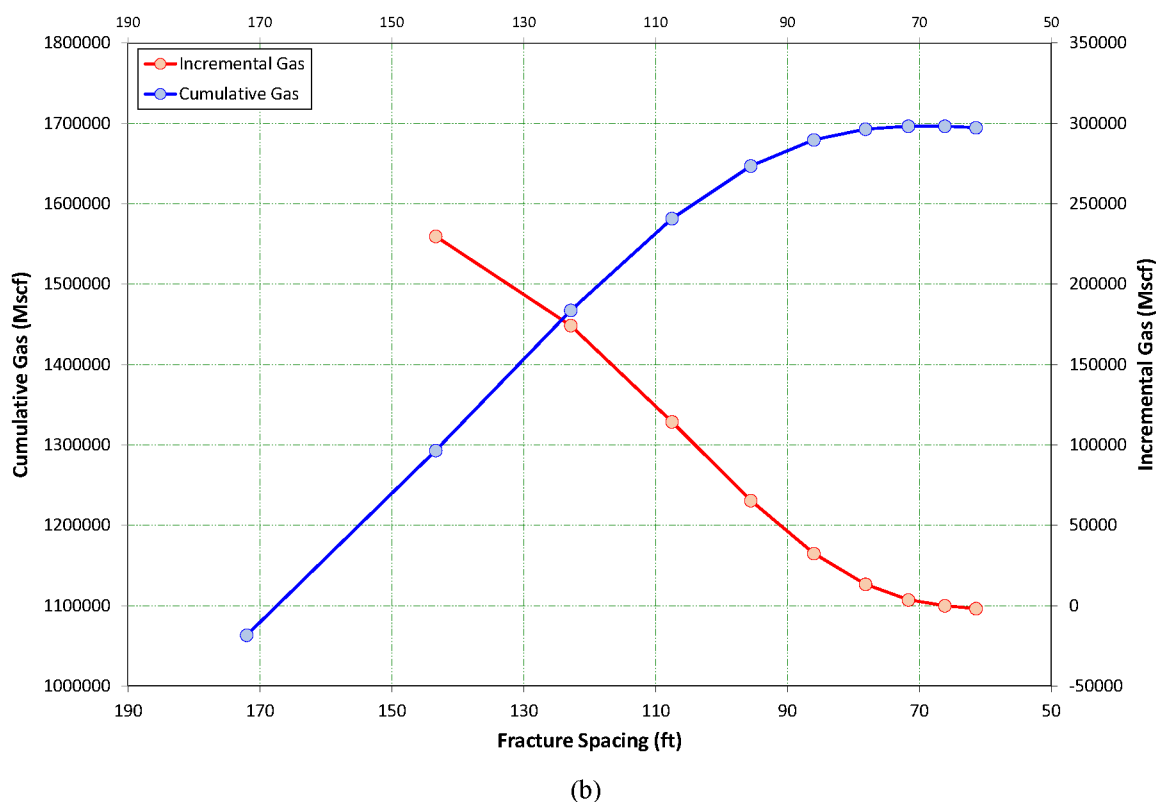


Figure 10—(a) Cumulative gas production for different fracture spacing; (b) Cumulative gas production and incremental gas production versus fracture spacing

Example 2: Interference from Infill Well

A common production practice in unconventional reservoirs is the multi-well pad production scheme where MFHW wells (parent wells) are drilled from the same surface location. Infill wells (child wells) may be drilled to access unswept resource located between the parent wells. This example demonstrates the performance impact suffered by a parent well in the presence of an active child well. Well-1 commences production at time zero. At its midlife, Well-2 is completed and brought onstream. Both wells are of similar completion (Fig. 11); they are traversed by infinite-conductivity hydraulic fractures fully penetrating an infinite reservoir. The reservoir is oil-bearing, has a constant pay and is bounded by no-flow top and bottom boundaries. Table 5 details the reservoir properties, oil properties, and well specification.

Table 5—Reservoir properties, oil properties, and well specification for Example 2

Reservoir Properties	k (md)	0.1
	θ (%)	8
	c_f (psi ⁻¹)	3.0E-6
	h (ft)	300
Oil Properties	Same as Table 2	
Well Specification	r_w (ft)	0.25
	L_w (ft)	5000
	z_w (ft)	75
	N_f	32
	x_f (ft)	250
	h_f (ft)	300

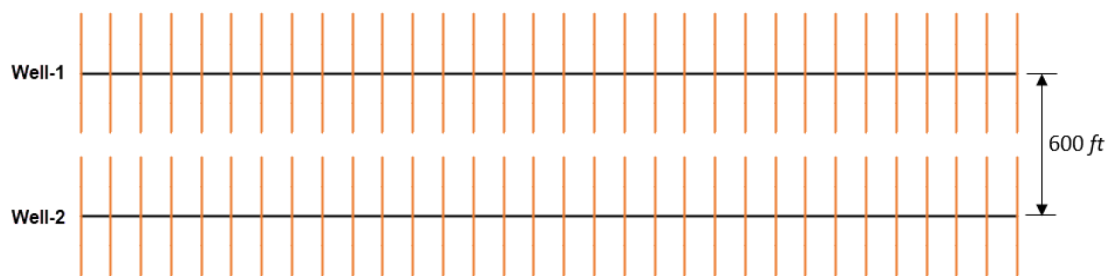
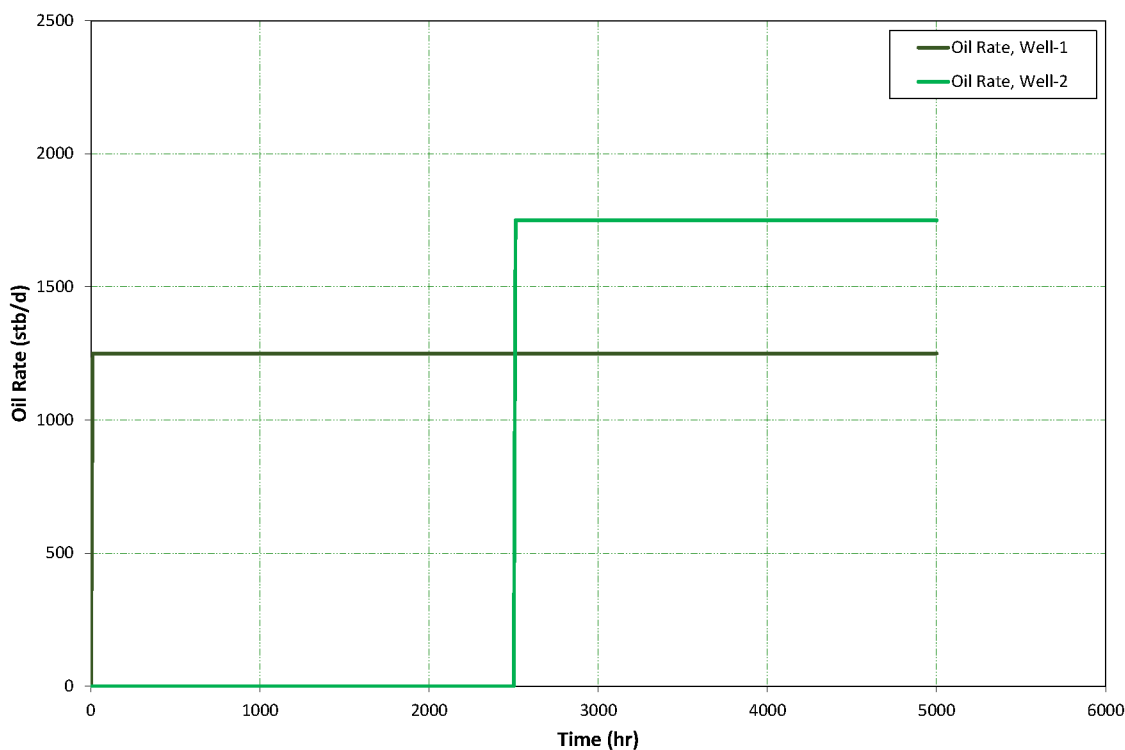


Figure 11—Three MFHW wells placed in a wine-rack pattern in a constant-pay, infinite reservoir with no-flow top and bottom boundaries

The production control of Well-1 and of Well-2 are as shown in Fig. 12a, their corresponding bottomhole pressure response (as solid lines) in Fig. 12b. Although Well-2 is brought online at the midlife of Well-1, its local pressure decline in the period prior confirms the spatiotemporal influence of Well-1's drawdown. The eventual production by Well-2, as expected, directly impacts Well-1, causing an inflexion in the hitherto "undisturbed" bottomhole pressure profile of Well-1.



(a)

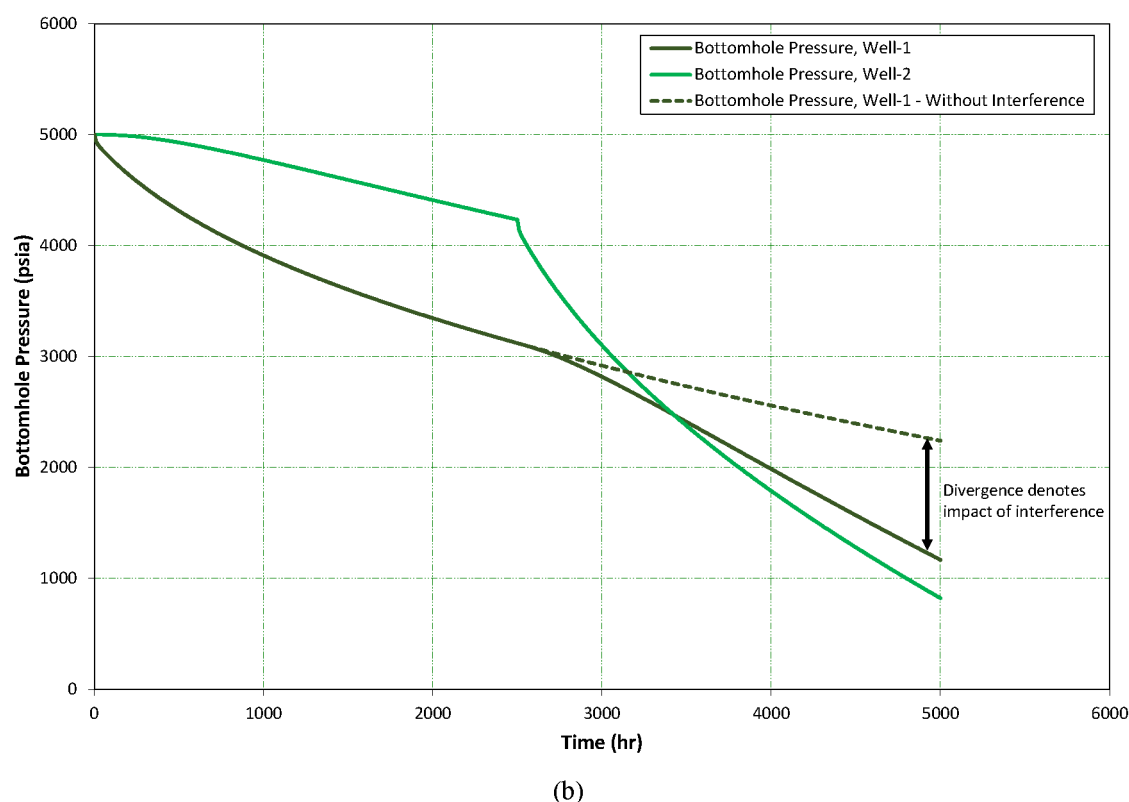


Figure 12—(a) Variable gas flow rate controlling well production;
(b) Bottomhole pressure at well due to variable flow rate control

For completeness, we also present the bottomhole pressure response (the dash line) of Well-1 if there was no interference, i.e., supposing Well-2 remained unoperated the entire time. This is plotted in Fig. 12b as a dash line. The impact of interference manifests in the late life of Well-1 as the divergence noted between the well's pressure profiles, with and without interference.

Example 3: History Matching and Forecasting

This example epitomizes possible field scenarios where a hydraulically fractured well with a complex lateral is drilled in an unconventional reservoir. The reservoir is oil-bearing (same fluid properties in Table 2); its porosity and rock compressibility are 20% and $3.0\text{E-}6 \text{ psi}^{-1}$ respectively. The well produces from 200-ft pay at a variable flow rate for the first 420 hours, after which it is shut in for 180 hours to monitor pressure build-up (see Fig. 13).

To obtain estimates of fracture properties, pressure transient analysis is performed using the well's production and buildup data. Fig. 14 is the diagnostic log-log plot showing the rate-normalized pressure-difference and semi-log-derivative rendition of both the drawdown and the buildup data. While the buildup data plot is well-behaved, the drawdown counterpart is relatively erratic owing to the changing flow rate. Using the specialized flow analysis, initial estimates of permeability and fracture half-length are obtained from the radial-flow line and the linear-flow line as 0.282 md and 123.804 ft respectively.

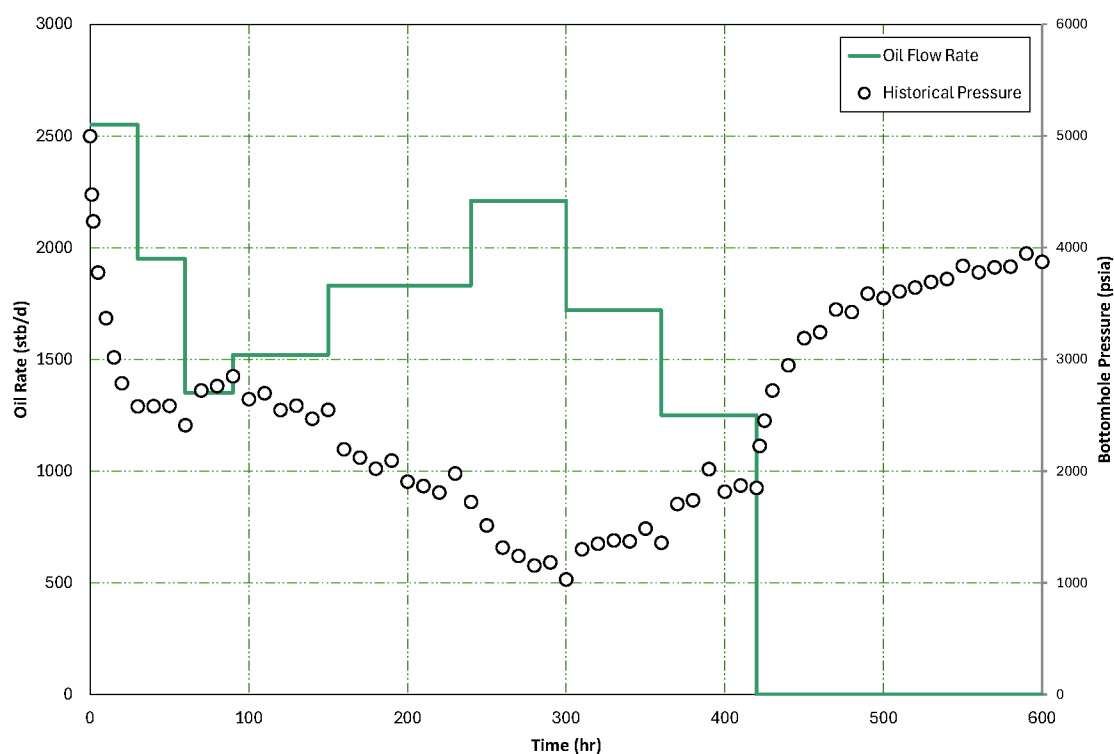


Figure 13—Historical data of well, showing variable production rate culminating in a shut-in; bottomhole flowing and shut-in pressure data are

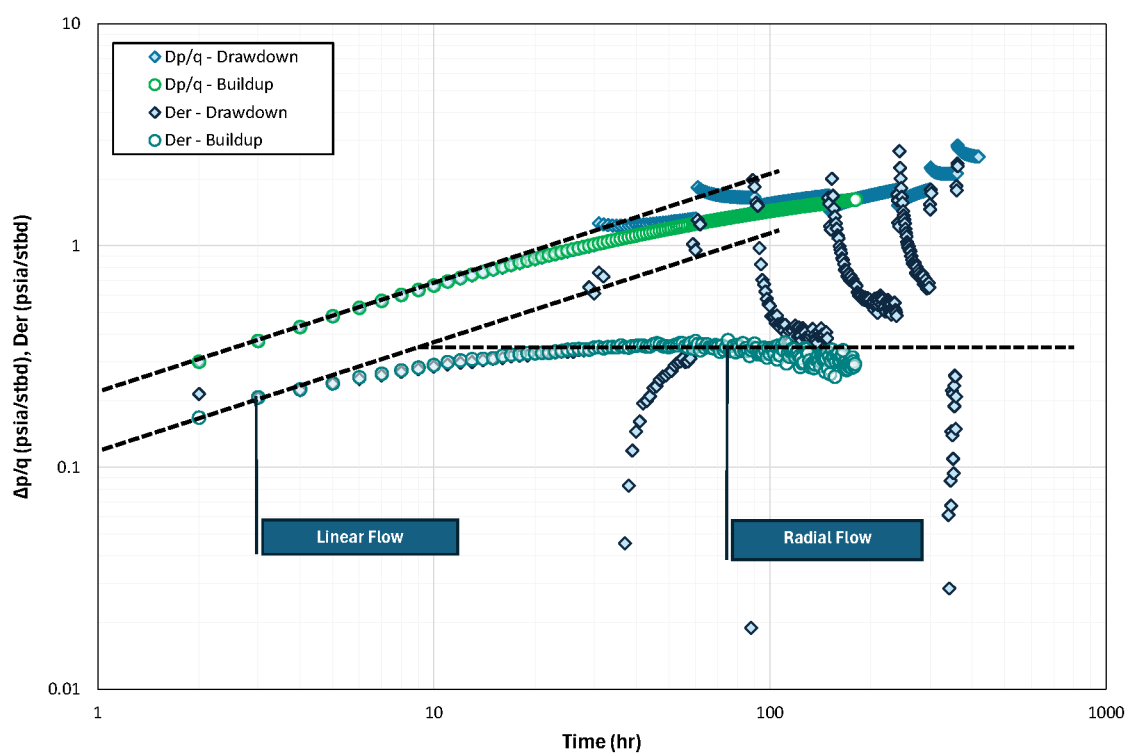


Figure 14—Pressure transient analysis of drawdown and buildup response on a log-log diagnostic plot

The above estimates of permeability and fracture half-length are then populated in our semi-analytical model. With the actual geometry of the well lateral specified (see vertical cross-section in Fig. 15), the model is fine-tuned to obtain the history match shown in the first 600 hours of Fig. 16. This resolves the permeability and fracture half-length inputs as 0.01 md and 125.01 ft respectively, based on completion data stipulating fracture spacing and height as 217 ft and 50 ft respectively. Afterward, the well's planned operating strategy is applied in the model to forecast future well performance, as captured in the subsequent 600 hours in Fig. 16.

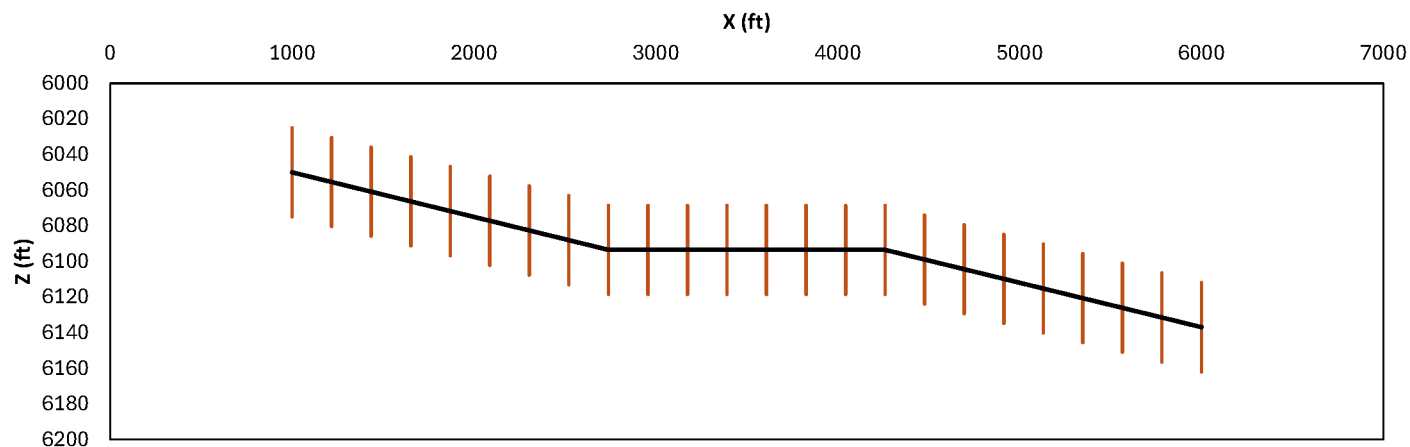


Figure 15—Vertical cross-section through the reservoir pay showing the well lateral intercepted by hydraulic fractures

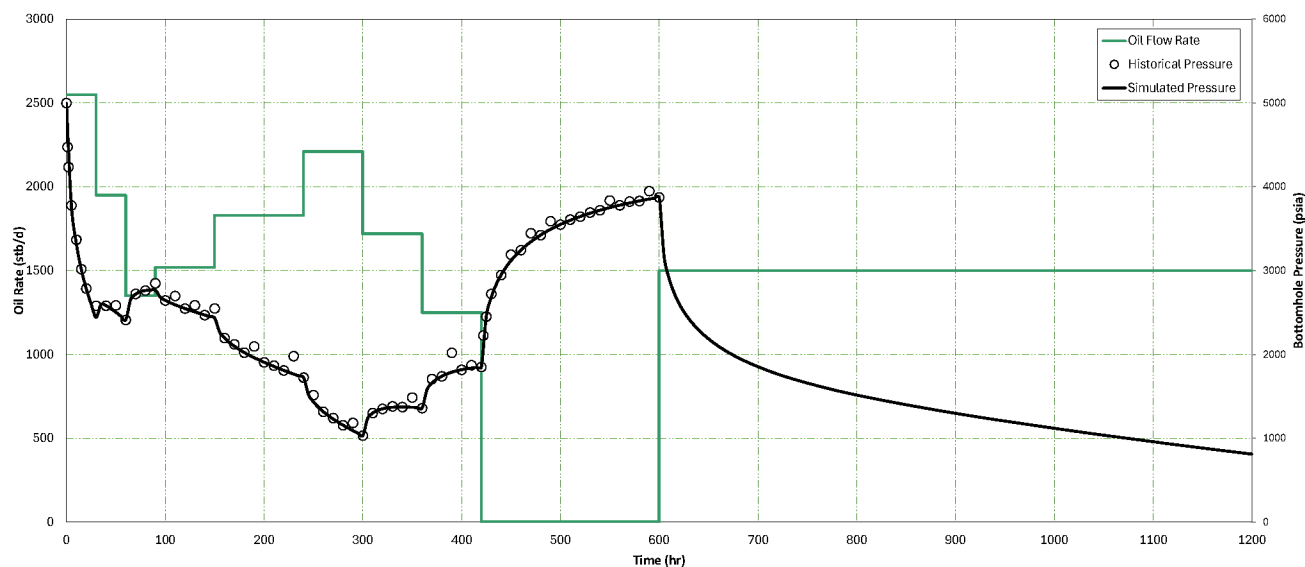


Figure 16—History match of the well's past performance plus pressure forecast based on planned operating strategy

Conclusions

In this paper, we develop a semi-analytical solution which finds utility in the assessment of multi-well performance and interference in unconventional reservoirs. The solution rigorously combines the boundary element method with an appropriate Green's function, such that arbitrary-shaped and infinite reservoirs can be modeled conveniently. Wells are controlled either by a specified flow rate or a specified pressure. Below are notable conclusions amongst others.

- Our solution differs from those existing in literature and industry in that complex laterals can be represented conveniently, unlike analytical models which are restricted to regular reservoir-well architectures and numerical solution limited by extreme gridding requirements. This flexibility gives our solution the capability for simulating unconventional resource development driven by irregular well patterns.
- The verification cases presented showcase the high, near-analytical accuracy possible with our solution. They cover a variety of complex reservoir-well scenarios, including fully and partially penetrating hydraulic fractures, uniform-flux and infinite- conductivity fractures, and irregular hydraulic fractures. The excellent agreement with results from a standard industry software confirms the robustness of our solution.
- We highlight the impact of source discretization on results accuracy. Lengthwise discretization is sufficient for representing fully penetrating fractures. Bidirectional discretization is necessary to achieve acceptable accuracy for partially penetrating fractures, especially when the fracture clearance from the reservoir top and bottom boundaries is significant.
- Three use-case examples demonstrate practical application of our solution to industry problems, including, but not limited to, fracture spacing, interference modeling, history matching and parameter estimation, and performance forecasting. We handle multi-well flow dynamics rigorously by use of superposition in space.
- The high degree of accuracy of our solution makes it useful as a calibration standard for optimizing grids in numerical simulation. Given its capability, it can also be deployed in synchronization with and as a reflection of actual field development for unconventional reservoirs producing single-phase oil or gas.

Nomenclature

B	=	formation volume factor, <i>bbl/stb</i>
c_t	=	total compressibility, <i>psi⁻¹</i>
Der	=	Bourdet/semi-log derivative, <i>psia/stb</i>
G	=	Green's function
h	=	reservoir thickness, <i>ft</i>
k	=	permeability, <i>md</i>
ℓ	=	reference length, <i>ft</i>
$\hat{\mathbf{n}}$	=	outward unit normal vector
N_f	=	number of hydraulic fractures
p	=	pressure, <i>psia</i>
p_i	=	initial pressure, <i>psi</i>
p_{wf}	=	bottom-hole flowing pressure, <i>psi</i>
q	=	flow rate, <i>stb/d</i>
r	=	radial distance, <i>ft</i>
s	=	Laplace domain parameter
S	=	Source function
t	=	time, <i>hr</i>
T	=	temperature, <i>°R</i>
$\mathbf{x}$	=	coordinate position (<i>x, y, z</i>)
ϕ	=	porosity, <i>fraction</i>
γ	=	fluid gravity, <i>dimensionless</i>
μ	=	viscosity, <i>cp</i>
η	=	diffusivity
δ	=	Dirac / Kronecker delta
∇	=	nabla / del operator
π	=	Pi constant
τ	=	time at start of fluid withdrawal, <i>hr</i>
Ω	=	domain
Γ	=	surface / boundary

Subscripts

D	=	dimensionless
f	=	hydraulic fracture
e	=	reservoir exterior
g	=	gas
i	=	initial
j	=	<i>j</i> -th reservoir region
o	=	oil
ref	=	reference
w	=	well
ξ	=	<i>x</i> -, <i>y</i> - or <i>z</i> -coordinate

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Appendix A - Dimensionless Variables

The dimensionless quantities appearing in Eq. 2 and in the rest of this paper are as defined below

$$p_D = 1.127 \times 10^{-3} \frac{2\pi k_{ref} h}{q_{ref} B\mu} [p_i - p(\mathbf{x}, t)] \quad (\text{A-1})$$

$$\tilde{q}_{wD}(\mathbf{x}_D, t_D) = \frac{\tilde{q}_w(\mathbf{x}, t)}{q_{ref}} \quad (\text{A-2})$$

$$t_D = \frac{\eta_{ref}}{\ell^2} t \quad (\text{A-3})$$

$$\xi_D = \frac{\xi}{\ell} \sqrt{\frac{k_{ref}}{k_\xi}} \quad (\xi = x, y, z) \quad (\text{A-4})$$

$$\eta_D = \frac{\eta}{\eta_{ref}} \quad (\text{A-5})$$

$$\eta = 2.637 \times 10^{-4} \frac{k_{ref}}{\phi \mu c_t} \quad (\text{A-6})$$

$$\eta_{ref} = 2.637 \times 10^{-4} \left(\frac{k}{\phi \mu c_t} \right)_{ref} \quad (\text{A-7})$$

$$k_{ref} = (k_x k_y k_z)^{1/3} \quad (\text{A-8})$$

The "ref" subscript marker connotes a reference quantity for normalizing and comparing similar variables.

Appendix B - Green's Function for a Point Source in an Infinite Slab Reservoir

The source function adopted in this study is obtained by use of the Newman product method (Newman 1936). The Newman product method derives the Green's function for a higher dimension (2D or 3D) from the product of 1D and/or 2D Green's functions. For an infinite slab reservoir, the defining point source can be decomposed into an infinite, vertical line source in an unbounded reservoir and an infinite, horizontal planar source in an infinite, horizontal slab reservoir. The decomposition is shown graphically in Fig. B-1.

In Laplace space, the Green's function describing 2D x - y radial flow to an infinite, vertical line source in an infinite reservoir (Figure B-1(b)) is given by Eq. B-1 (Idorenyin and Shirif, 2015, 2016, 2018):

$$\bar{G}_{xy}(x_D, y_D, x_{wD}, y_{wD}; s) = \frac{1}{2\pi\eta_D} K_0\left(r_D \sqrt{\frac{s}{\eta_D}}\right) \quad (\text{B-1})$$

where

$$r_D = \sqrt{(x_D - x_{wD})^2 + (y_D - y_{wD})^2} \quad (\text{B-2})$$

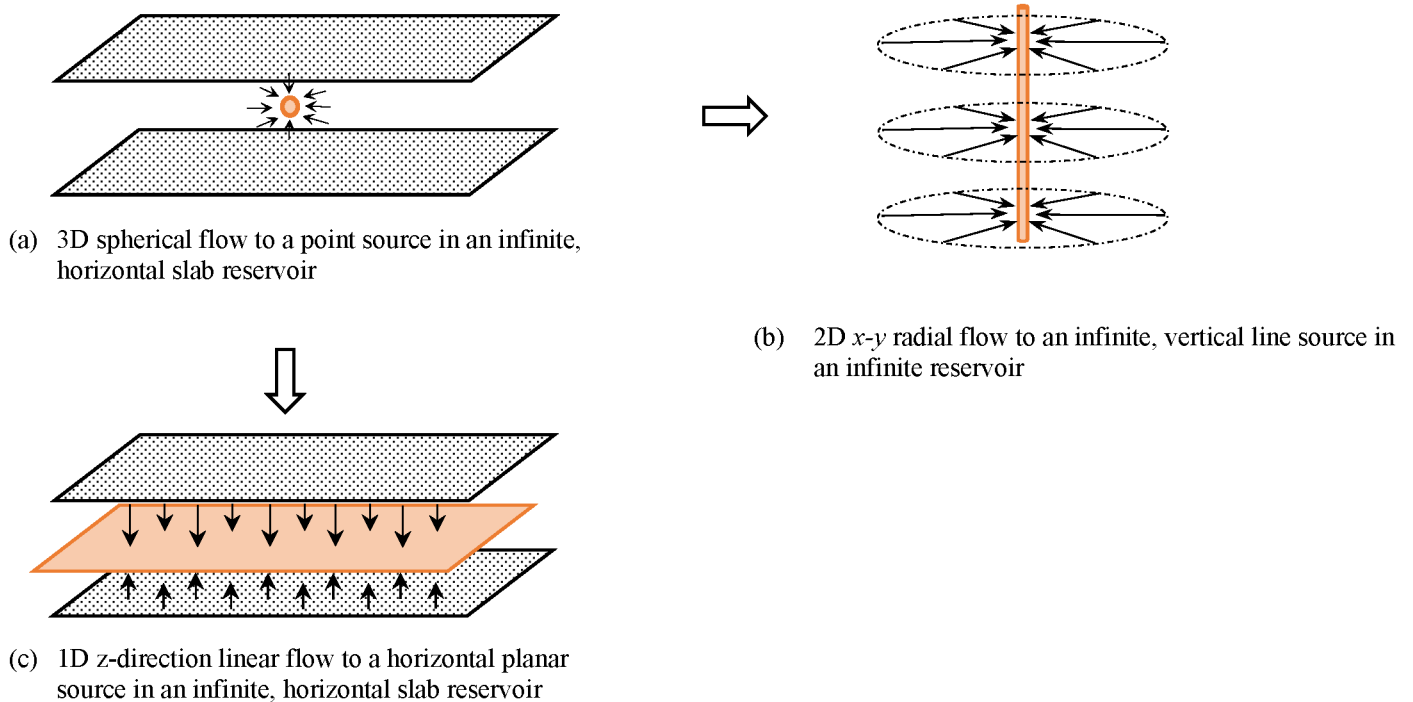


Figure B-1—Decomposition of a point source in (a) an infinite slab reservoir into constituent sources: (b) a vertical line source in an unbounded domain, and (c) a horizontal planar source in a slab flow domain

On the other hand, the Green's function for 1D z -direction linear flow to an infinite, horizontal planar source in an infinite, horizontal slab reservoir (Figure B-1(c)) is given by (Idorenyin and Shirif, 2015)

$$\bar{G}_z(z_D, z_{wD}; s) = \frac{1}{2\eta_D \sqrt{\frac{s}{\eta_D}}} \left[\frac{e^{-\sqrt{\frac{s}{\eta_D}} |z_D - z_{wD}|} + e^{-\sqrt{\frac{s}{\eta_D}} (z_D + z_{wD})} + e^{-\sqrt{\frac{s}{\eta_D}} \{2z_{eD} - (z_D + z_{wD})\}} + e^{-\sqrt{\frac{s}{\eta_D}} \{2z_{eD} - |z_D - z_{wD}|\}}}{1 - e^{-2\sqrt{\frac{s}{\eta_D}} z_{eD}}} \right]$$

which, in terms of hyperbolic functions, becomes

$$\bar{G}_z(z_D, z_{wD}; s) = \frac{1}{2\eta_D \sqrt{\frac{s}{\eta_D}}} \left[\frac{\cosh\left[\sqrt{\frac{s}{\eta_D}} (z_{eD} - |z_D - z_{wD}|)\right] + \cosh\left[\sqrt{\frac{s}{\eta_D}} \{z_{eD} - (z_D + z_{wD})\}\right]}{\sinh\left(z_{eD} \sqrt{\frac{s}{\eta_D}}\right)} \right] \quad (\text{B-3})$$

Using Newman's product method, the source function for a point source in an infinite, horizontal slab reservoir (Figure B-1(a)) is the product of the constituent source functions. That is,

$$G(\mathbf{x}_D, \mathbf{x}_{wD}; t_D - \tau_D) = G_{xy}(x_D, y_D, x_{wD}, y_{wD}; t_D - \tau_D) \cdot G_z(z_D, z_{wD}; t_D - \tau_D) \quad (\text{B-4})$$

In Laplace space, the product in Eq. B-4 is expressed as,

$$\bar{G}(\mathbf{x}_D, \mathbf{x}_{wD}; s) = \mathcal{L}\{G_{xy}(x_D, y_D, x_{wD}, y_{wD}; t_D - \tau_D) \cdot G_z(z_D, z_{wD}; t_D - \tau_D)\}$$

which, using complex integration, becomes

$$\bar{G}(\mathbf{x}_D, \mathbf{x}_{wD}; s) = \frac{1}{2\pi i} \int_{\text{Br}_1} \bar{G}_z(z_D, z_{wD}; s) \cdot \bar{G}_{xy}(x_D, y_D, x_{wD}, y_{wD}; s - s_n) ds \quad (\text{B-5})$$

where $i = \sqrt{-1}$ and Br_1 is a Bromwich contour defined such that all singular points s_n of $\bar{G}_{xy}(x_D, y_D, x_{wD}, y_{wD}; s - s_n)$ are enclosed within it. Note that the two Green's function expressions under the integral sign in Eq. B-5 are given by Eq. B-1 and B-3. Using the residue theorem, Eq. B-5 is evaluated as

$$\bar{G}(\mathbf{x}_D, \mathbf{x}_{wD}; s) = \sum_n \text{Res} \left[\bar{G}_z(z_D, z_{wD}; s) \cdot \bar{G}_{xy}(x_D, y_D, x_{wD}, y_{wD}; s - s_n) \right] \quad (\text{B-6})$$

The singularities of $\bar{G}_z(z_D, z_{wD}; s)$ occur at the points where

$$\sqrt{\frac{s_n}{\eta_D}} = \frac{in\pi}{z_{eD}} \quad (n = 0, \pm 1, \pm 2 \dots) \quad (\text{B-7})$$

Evaluating the residue in Eq. B-6 while considering the singularities stated in Eq. B-7, it can be shown that Eq. B-6 simplifies as

$$\bar{G}(\mathbf{x}_D, \mathbf{x}_{wD}; s) = \frac{1}{2\pi\eta_D z_{eD}} \left[K_0 \left(r_D \sqrt{\frac{s}{\eta_D}} \right) + 2 \sum_{n=1}^{\infty} \cos \left(\frac{n\pi z_D}{z_{eD}} \right) \cos \left(\frac{n\pi z_{wD}}{z_{eD}} \right) \cdot K_0 \left(r_D \sqrt{\frac{s}{\eta_D} + \left(\frac{n\pi}{z_{eD}} \right)^2} \right) \right] \quad (\text{B-8})$$

Eq. B-8 is the Green's function of a point source in an infinite slab reservoir. It is equivalent to that reported by Ozkan and Raghavan (1991) with the diffusivity ratio set to unity, that is, $\eta_D = 1$. It's derivative with respect to r_D is given by:

$$\frac{\partial \bar{G}}{\partial r_D} = \frac{-1}{2\pi\eta_D z_{eD}} \left[\sqrt{\frac{s}{\eta_D}} K_1 \left(r_D \sqrt{\frac{s}{\eta_D}} \right) + 2 \sum_{n=1}^{\infty} \cos \left(\frac{n\pi z_D}{z_{eD}} \right) \cos \left(\frac{n\pi z_{wD}}{z_{eD}} \right) \cdot \sqrt{\frac{s}{\eta_D} + \left(\frac{n\pi}{z_{eD}} \right)^2} K_1 \left(r_D \sqrt{\frac{s}{\eta_D} + \left(\frac{n\pi}{z_{eD}} \right)^2} \right) \right] \quad (\text{B-9})$$

Appendix C - BEM Framework

In Laplace space, the Green's function solution of the diffusivity equation, for flow of a slightly compressible fluid in a homogeneous porous domain Ω bounded by the surface Γ , as shown in Fig. 1, is given by Eq. 2 above. We focus on flow in a slab reservoir in this study; the attendant Green's function and source function for this type of flow domain, in Laplace domain, are given by Eq. B-8 and Eq. 3 respectively. For computer implementation of Eq. 2, we resort to the boundary element method. This requires discretizing the reservoir boundary into planar cells (called elements) $j = 1, 2, \dots, N$, as shown in Fig. 1. In this study, we adopt quadrilateral boundary elements. For the discretized reservoir, Eq. 2 becomes

$$\bar{p}(\mathbf{x}, s) = 2\pi\eta \sum_{\ell=1}^M \bar{q}_{w\ell} \bar{S} + \eta \sum_{j=1}^N \int_{\Gamma_j} (\bar{G} \nabla \bar{p}_j - \bar{p}_j \nabla \bar{G}) \cdot \hat{\mathbf{n}} d\Gamma' \quad (\text{C-1})$$

Note that, for convenience, the arguments and the dimensionless subscript marker "D" have been dropped in Eq. C-1. Here we use the isoparametric BEM implementation (Kikani and Horne 1989, 1993); that is, the boundary quantities (pressure and flux) are approximated as bilinear functions over quadrilateral boundary elements (which are also represented with bilinear functions). Since each boundary element is characterized by four nodes, Eq. B-1 thus becomes

$$\bar{p}(\mathbf{x}, s) = 2\pi\eta \sum_{\ell=1}^M \bar{q}_{w\ell} \bar{S} + \eta \sum_{j=1}^N \sum_{n=1}^4 \int_{\Gamma_j} N_n \bar{G} \nabla \bar{p}_{jn} \cdot \hat{\mathbf{n}} d\Gamma' - \eta \sum_{j=1}^N \sum_{n=1}^4 \int_{\Gamma_j} N_n \bar{p}_{jn} \nabla \bar{G} \cdot \hat{\mathbf{n}} d\Gamma' \quad (\text{C-2})$$

where $N_n(\xi, \zeta)$, is the bilinear shape function which, for the nodes of a quadrilateral boundary element, is given by

$$N_n(\xi, \zeta) = \frac{1}{4} (1 + \xi_n \xi) (1 + \zeta_n \zeta) \quad (\text{C-3})$$

The values of ξ_n and ζ_n along the intrinsic coordinates (ξ, ζ) are as given in Table C-1 below.

Table C-1—Intrinsic coordinates of boundary-element nodes

n	ξ_n	ζ_n
1	-1.0	-1.0
2	1.0	-1.0
3	1.0	1.0
4	-1.0	1.0

By use of a shape function, the global coordinate $\mathbf{x} = (x, y, z)$ is defined in terms two orthogonal local coordinates $\xi \in (-1, +1)$ and $\zeta \in (-1, +1)$. The local coordinate system is attached to the boundary element as shown in Fig. C-1. At the boundary nodes, the boundary quantities are constant; hence they are pulled out of the integral, resulting in

$$\bar{p}(\mathbf{x}, s) + \eta \sum_{j=1}^N \sum_{n=1}^4 \bar{p}_{jn} \int_{\Gamma_j} N_n \nabla \bar{G} \cdot \hat{\mathbf{n}} d\Gamma' - \eta \sum_{j=1}^N \sum_{n=1}^4 \nabla \bar{p}_{jn} \cdot \hat{\mathbf{n}} \int_{\Gamma_j} N_n \bar{G} d\Gamma' - 2\pi\eta \sum_{\ell=1}^M \bar{q}_{w\ell} \bar{S} = 0 \quad (\text{B-4})$$

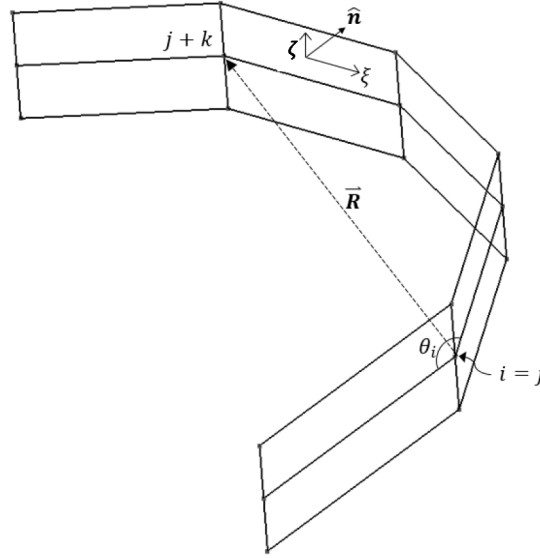


Figure C-1—Transformation of global coordinates (x,y,z) to local coordinates (ξ,ζ) attached to each boundary element

For a well-posed problem, boundary conditions must be specified - pressure and flux must be exclusively prescribed at the reservoir boundary Γ . Flux is prescribed on a section Γ_a containing N_a boundary elements, and pressure on the rest section Γ_b containing N_b boundary elements, such that $\Gamma_a \cup \Gamma_b = \Gamma$, $\Gamma_a \cap \Gamma_b = \emptyset$ and $N_a + N_b = N$. This means the pressure data on Γ_a is unknown and likewise the flux data on Γ_b , making a total of N boundary data that needs to be determined. Determining these unknown boundary data requires a system of N equations. The equations are obtained by applying the collocation procedure on the boundary nodes ($i = 1, 2, \dots, N$), yielding

$$\frac{\theta_i}{2\pi} \bar{p}_i(\mathbf{x}_b, s) + \eta \sum_{j=1}^N \sum_{n=1}^4 \bar{p}_{jn} \int_{\Gamma_j} N_n \nabla \bar{G} \cdot \hat{\mathbf{n}} d\Gamma' - \eta \sum_{j=1}^N \sum_{n=1}^4 \nabla \bar{p}_{jn} \cdot \hat{\mathbf{n}} \int_{\Gamma_j} N_n \bar{G} d\Gamma' - 2\pi\eta \sum_{\ell=1}^M \bar{q}_{w\ell} \bar{S} = 0 \quad (i = 1, 2, \dots, N) \quad (\text{C-5})$$

where θ_i is the angle enclosed by the boundary geometry at the i -th node, as shown in Fig. C-1. At points interior to the reservoir, $\theta = 2\pi$. Setting,

$$A_{ij}^n = \eta \int_{\Gamma_j} N_n \nabla \bar{G} \cdot \hat{\mathbf{n}} d\Gamma' \quad (\text{C-6})$$

$$B_{ij}^n = -\eta \int_{\Gamma_j} N_n \bar{G} d\Gamma' \quad (\text{C-7})$$

So That,

$$A_{ij} = \frac{\theta_i}{2\pi} \delta_{ij} + \eta \sum_{e_j \in \mathbf{x}_i} \int_{\Gamma_j} N_n \nabla \bar{G} \cdot \hat{\mathbf{n}} d\Gamma' \quad (\text{C-8})$$

$$B_{ij} = -\eta \sum_{e_j \in \mathbf{x}_i} \int_{\Gamma_j} N_n \bar{G} d\Gamma' \quad (\text{C-9})$$

$$C_{i\ell} = -2\pi\eta \bar{S}(\mathbf{x}_b, \mathbf{x}_\ell; s) \quad (\text{C-10})$$

$$\bar{q}_j = \nabla \bar{p}_j \cdot \hat{\mathbf{n}} \quad (\text{C-11})$$

where $e_j \in \mathbf{x}_i$ denotes element j abutting node i . The Kronecker delta is defined as

$$\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} \quad (\text{C-12})$$

and $\bar{q}_j$ is flux across the j -th boundary element caused by the outward normal pressure gradient $\nabla \bar{p}_j \cdot \hat{\mathbf{n}}$ acting at the boundary element. We compute the surface integral present in each of Eq. C-6 and Eq. C-7 numerically using Gauss-Legendre quadrature (Abramowitz and Stegun 1965). Substituting Eq. C-8 through C-11 into Eq. C-5, yields

$$\sum_{j=1}^N A_{ij} \bar{p}_j + \sum_{j=1}^N B_{ij} \bar{q}_j + \sum_{\ell=1}^M C_{i\ell} \bar{q}_{w\ell} = 0 \quad (i = 1, 2, \dots, N) \quad (\text{C-13})$$

Eq. C-13 can be written succinctly in matrix form as

$$\mathbf{A}\bar{\mathbf{p}} + \mathbf{B}\bar{\mathbf{q}} + \mathbf{C}\bar{\mathbf{q}}_w = \mathbf{0} \quad (\text{C-14})$$

where $\mathbf{A}$ and $\mathbf{B}$ are $N \times N$ matrices, $\mathbf{C}$ is an $N \times M$ matrix, $\bar{\mathbf{p}}$ and $\bar{\mathbf{q}}$ are N -element vectors, $\bar{\mathbf{q}}_w$ is an M -element vector and $\mathbf{0}$ is an N -element zero vector. Eq. C-14 is a system of linear equations which, when solved, determines the boundary quantities (pressure and/or flux, depending on what boundary condition had been prescribed). It is a concise statement of the transient response in the reservoir being the combined effects of the boundary pressure, the boundary flux, and the well flow rate.